Ramucirumab plus paclitaxel versus placebo plus paclitaxel in patients with previously treated advanced gastric or gastro-oesophageal junction adenocarcinoma (RAINBOW): a double-blind, randomised phase 3 trial

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Summary

Background VEGFR-2 has a role in gastric cancer pathogenesis and progression. We assessed whether ramucirumab, a monoclonal antibody VEGFR-2 antagonist, in combination with paclitaxel would increase overall survival in patients previously treated for advanced gastric cancer compared with placebo plus paclitaxel.

Methods This randomised, placebo-controlled, double-blind, phase 3 trial was done at 170 centres in 27 countries in North and South America, Europe, Asia, and Australia. Patients aged 18 years or older with advanced gastric or gastro-oesophageal junction adenocarcinoma and disease progression on or within 4 months after first-line chemotherapy (platinum plus fluoropyrimidine with or without an anthracycline) were randomly assigned with a centralised interactive voice or web-response system in a 1:1 ratio to receive ramucirumab 8 mg/kg or placebo intravenously on days 1 and 15, plus paclitaxel 80 mg/m^2 intravenously on days 1, 8, and 15 of a 28-day cycle. A permuted block randomisation, stratified by geographic region, time to progression on first-line therapy, and disease measurability, was used. The primary endpoint was overall survival. Efficacy analysis was by intention to treat, and safety analysis included all patients who received at least one treatment with study drug. This trial is registered with ClinicalTrials.gov, number NCT01170663, and has been completed; patients who are still receiving treatment are in the extension phase.

Findings Between Dec 23, 2010, and Sept 23, 2012, 665 patients were randomly assigned to treatment—330 to ramucirumab plus paclitaxel and 335 to placebo plus paclitaxel. Overall survival was significantly longer in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group (median 9·6 months [95% CI 8·5–10·8] *vs* 7·4 months [95% CI 6·3–8·4], hazard ratio 0·807 [95% CI 0·678–0·962]; p=0·017). Grade 3 or higher adverse events that occurred in more than 5% of patients in the ramucirumab plus paclitaxel group versus placebo plus paclitaxel included neutropenia (133 [41%] of 327 *vs* 62 [19%] of 329), leucopenia (57 [17%] *vs* 22 [7%]), hypertension (46 [14%] *vs* eight [2%]), fatigue (39 [12%] *vs* 18 [5%]), anaemia (30 [9%] *vs* 34 [10%]), and abdominal pain (20 [6%] *vs* 11 [3%]). The incidence of grade 3 or higher febrile neutropenia was low in both groups (ten [3%] *vs* eight [2%]).

Interpretation The combination of ramucirumab with paclitaxel significantly increases overall survival compared with placebo plus paclitaxel, and could be regarded as a new standard second-line treatment for patients with advanced gastric cancer.

Funding Eli Lilly and Company.

Introduction

Gastric cancer is the fifth most common malignancy, and the third leading cause of cancer mortality worldwide.[1] Currently, platinum-based and fluoropyrimidine-based combinations are accepted worldwide as established first-line drug regimens.[2] There are not many treatment options after failure of first-line therapy. In randomised trials, selected second-line chemotherapy significantly improved overall survival compared with best supportive care,[3–5] however, median survival was less than 6 months. Therefore, new, more active second-line treatment options are needed.

VEGF and VEGFR-2-mediated signalling and angiogenesis contribute to the pathogenesis of gastric cancer. In patients with gastric cancer, circulating VEGF levels are associated with increased tumour aggressiveness and reduced survival.[6,7] In animal models of gastric adenocarcinoma, VEGFR-2 inhibition reduced tumour growth and vascularity.[8] First-line treatment with bevacizumab, a VEGF-A-directed monoclonal antibody, in combination with chemotherapy was associated with significantly improved proportions of patients achieving an objective response and progression-free survival, and non-significantly improved overall survival in patients with metastatic gastric cancer.[9,10] Ramucirumab, a human IgG1 monoclonal antibody VEGFR-2 antagonist, prevents ligand binding and receptor-mediated pathway activation in endothelial cells.[11] Paclitaxel was chosen for the

combination based on single-agent second-line trials;[12–14] the results of a retrospective analysis in gastric cancer indicated similar efficacy between frequently used second-line drugs (taxanes or irinotecan).[15] Weekly paclitaxel is better tolerated and more efficacious than 3-weekly paclitaxel in metastatic breast cancer.[16] More recently, in a Japanese randomised trial, weekly paclitaxel was associated with a good toxicity profile compared with irinotecan as second-line therapy in patients with gastric cancer.[17]

We assessed the safety and efficacy of ramucirumab plus paclitaxel in patients with advanced gastric or gastro-oesophageal junction adenocarcinoma with disease progression after first-line combination chemotherapy.

Methods

Study design and patients

RAINBOW was a double-blind, placebo-controlled phase 3 trial. Eligibility criteria included age 18 years and older; having metastatic or non-resectable, locally advanced gastric or gastro-oesophageal junction adenocarcinoma; documented objective radiological or clinical disease progression during or within 4 months of the last dose of first-line platinum and fluoropyrimidine doublet with or without anthracycline; an Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1; and measureable or non-measurable evaluable disease (defined with Response Evaluation Criteria In Solid Tumors [RECIST], version 1.1).[18] Exclusion criteria included having squamous or undifferentiated gastric cancer; gastro-intestinal perforation, fistulae, or any arterial thrombo-embolic event within 6 months, or any significant gastrointestinal bleeding or any significant venous thromboembolism within 3 months before randomisation; or poorly controlled hypertension. The appendix provides the full inclusion and exclusion criteria.

Each centre's institutional review board or independent ethics committee approved the study. The trial followed the principles of the Declaration of Helsinki and the Good Clinical Practice Guidelines of the International Conference on Harmonisation. All patients provided written informed consent.

Randomisation and masking

Patients were randomly assigned in a 1:1 ratio to ramucirumab plus paclitaxel or placebo plus paclitaxel using a randomisation sequence generated using the permuted blocks method within each stratum by a statistician not involved in the study activities. Randomisation was stratified by geographic region (region 1, Europe, Israel, Australia, and the USA; region 2, Argentina, Brazil, Chile, and Mexico; and region 3, Japan, South Korea, Hong Kong, Singapore, and Taiwan), time to progression after first dose of first-line therapy (<6 months *vs* ≥6 months), and disease measurability (measurable *vs* non-measurable). This sequence was programmed into a centralised interactive voice or web-response system. Study sites enrolled patients by accessing the centralised interactive voice or web-response system. The interactive voice or web-response system then assigned a unique identification number to each patient, and randomly assigned patients to one of the two treatment groups.

Patients, medical staff, study investigators, individuals who handled and analysed the data, and the funder were masked to treatment assignment. Ramucirumab and placebo for infusion were identical in appearance to preserve masking. Unmasking could be done for individual patients only on the request of a study physician in case knowledge of the identity of study drug was important for the treatment of serious adverse events.

Procedures

Patients received either ramucirumab 8 mg/kg (ImClone Systems, Branchburg, NJ, USA) or placebo intravenously on days 1 and 15, plus paclitaxel 80 mg/m² intravenously on days 1, 8, and 15 of a 28-day cycle. Patients received study treatment until disease progression, unacceptable toxicity, or withdrawal of consent. Crossover between treatment groups was not allowed. Criteria for discontinuation of patients from study treatment, and for dose modifications to manage treatment-related toxicities are presented in the appendix. All patients received supportive care if indicated.

CT scans were done every 6 weeks. Safety data were gathered continuously and local laboratory assessments (including haematology, clinical chemistry, coagulation, and urinalysis [appendix]) were done at baseline, before each treatment, at the end of therapy, and 30 days after the last dose of study drug; adverse events were graded in accordance with the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE; version 4.02).[19] We planned to assess quality of life every 6 weeks until progression using the European Organisation for Research and Treatment of Cancer quality-of-life questionnaire (EORTC QLQ-C30, version 3.0)[20] and the EuroQoL five-dimension, three-level health status questionnaire (EQ-5D-3L).[21] Performance status was assessed at the start of each cycle, at the end of therapy, and at the 30-day follow-up. Blood for analysis of anti-ramucirumab antibodies (immunogenicity) was obtained at baseline, day 15 of cycle 2 and day 1 of cycle 4, and at the 30-day follow-up, and patients' sera were analysed as detailed in the appendix.

Outcomes

The primary outcome was overall survival, defined as the time from randomisation to death from any cause. Secondary outcomes were progression-free survival, defined as time from randomisation to radiographic progression or death; objective tumour response, defined as the proportion of patients who had a best response of complete response or partial response; disease control, defined as the proportion of patients who had a best

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See **Online** for appendix

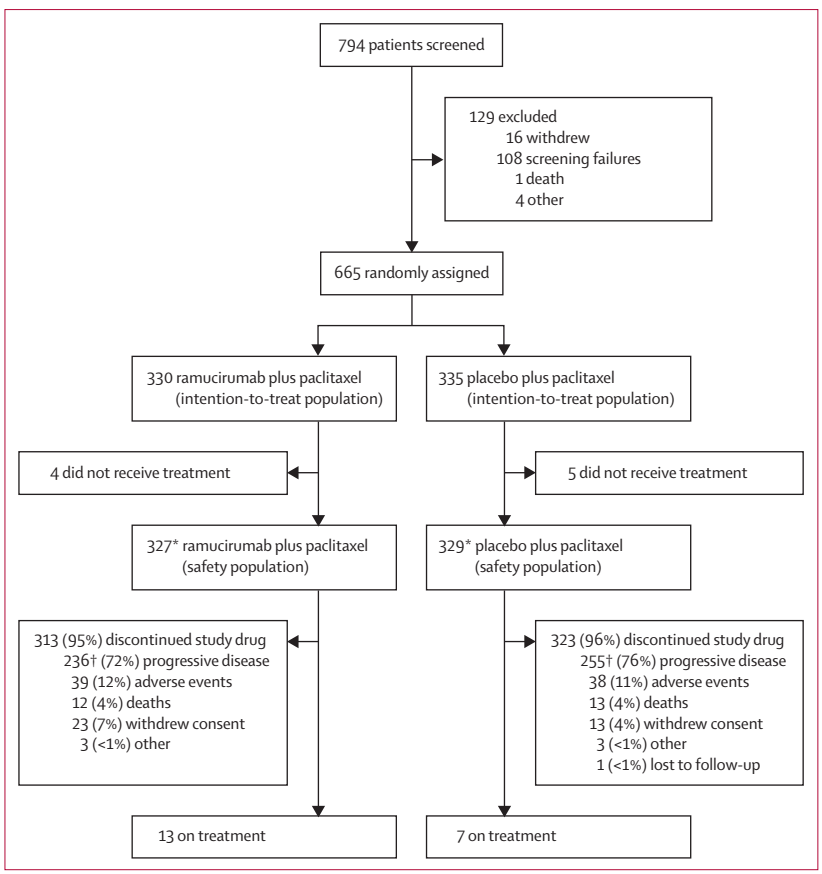

Figure 1: **Trial profile**
*One patient was randomly assigned to the placebo group, but received one dose of ramucirumab. †Radiographic progression or symptomatic deterioration.

response of complete response, partial response, or stable disease; patient-reported outcomes, as assessed using EORTC QLQ-C30 and EQ-5D-3L; immunogenicity of ramucirumab, where a treatment-emergent antibody-positive response was defined as a post-baseline positive response greater than four times increase in the antibody titre, or a missing baseline assessment and an on-treatment titre of at least 1:20; and safety. Disease progression and tumour response were assessed by investigators in accordance with RECIST 1.1.[18]

Statistical analysis

We calculated that to achieve 90% power to detect an overall difference in survival between the two treatment groups (hazard ratio [HR] 0·75; anticipated median overall survival 7·0 months in the placebo plus paclitaxel group *vs* 9·3 months in the ramucirumab plus paclitaxel group) with a one-sided α of 0·025 (two-sided 0·05), 510 deaths were needed, and 663 patients would need to be randomly assigned. The results presented here are based on a two-sided α of 0·05.

We used a log-rank test, stratified by geographic region, time to progression on first-line therapy, and disease measurability, to assess overall survival and progression-free survival. Estimations of time-to-event curves were generated with the Kaplan-Meier method. The HR was estimated with a stratified Cox proportional hazards model. We did a pre-planned multivariate analysis with a stepwise Cox regression model of predefined baseline characteristics to examine the effect of treatment on overall survival and progression-free survival after adjustment for significant prognostic factors. The proportion of patients achieving an objective response and disease control were compared between treatment groups with the Cochran-Mantel-Haenszel test, with adjustment for the randomisation strata. All analyses were done with SAS (version 9.2).

Efficacy analyses were based on the intention-to-treat population and predefined subgroups. The intention-to-treat population comprised all randomly assigned patients, irrespective of whether the patient received study medication. Safety analyses included all patients who received at least one dose of any study drug.

EORTC QLQ-C30 and EQ-5D data were scored according to developer guidelines and summarised descriptively for the intention-to-treat population.[22,23] The EQ-5D index score was calculated using the algorithm developed to represent UK population preferences for health states.[24]

This trial is registered with ClinicalTrials.gov, number NCT01170663.

Role of the funding source

The study funder provided the study drug and collaborated with investigators on protocol, study design, data gathering, analysis, and interpretation, and writing and preparation of this report. HW prepared the first draft in collaboration with the funder and other coauthors. HW had full access to all patient-level study data and all authors approved the submission for publication.

Results

Between Dec 23, 2010, and Sept 23, 2012, 665 (84%) of 794 screened patients were randomly assigned to receive ramucirumab plus paclitaxel (n=330) or placebo plus paclitaxel (n=335) at 170 centres in 27 countries in North and South America, Europe, Asia, and Australia (appendix). Figure 1 shows the trial profile. All patients were included in the efficacy analyses. As of data cutoff (July 12, 2013), with a median follow-up for overall survival of 7·9 months (IQR 4·2–13·0), 516 (78%) of 665 patients had died. 13 (4%) patients in the ramucirumab and paclitaxel group and seven (2%) in the placebo and paclitaxel group are still receiving treatment, and are in the extension phase of the study.

Baseline characteristics of patients and their tumours were generally well balanced between the groups (table 1). 662 of 665 patients received previous treatment with platinum-based and fluoropyrimidine-based chemotherapy regimens, including regimens with an anthracycline (163 [25%]); of the remaining

	Ramucirumab plus paclitaxel (n=330)	Placebo plus paclitaxel (n=335)
Age (years)		
Median (range)	61 (25–83)	61 (24–84)
<65	204 (62%)	212 (63%)
≥65	126 (38%)	123 (37%)
Sex		
Male	229 (69%)	243 (73%)
Ethnic origin*		
White	208 (63%)	199 (59%)
Asian	110 (33%)	121 (36%)
Black or other	12 (4%)	15 (4%)
ECOG performance status		
0	117 (35%)	144 (43%)
1	213 (65%)	191 (57%)
Geographic region†		
1	198 (60%)	200 (60%)
2	23 (7%)	21 (6%)
3	109 (33%)	114 (34%)
Site of primary tumour		
Gastric	264 (80%)	264 (79%)
Gastro-oesophageal junction adenocarcinoma	66 (20%)	71 (21%)
Disease‡		
Measurable	267 (81%)	273 (81%)
Non-measurable	63 (19%)	62 (19%)
Time to progressive disease on first-line therapy‡		
<6 months	250 (76%)	256 (76%)
≥6 months	80 (24%)	79 (24%)
Disease progression		
During first-line therapy	227 (69%)	217 (65%)
Tumor grade		
Well differentiated	28 (8%)	22 (7%)
Moderately differentiated	96 (29%)	106 (32%)
Poorly differentiated	186 (56%)	186 (56%)
Unknown or missing	20 (6%)	21 (6%)

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	Ramucirumab plus paclitaxel (n=330)	Placebo plus paclitaxel (n=335)
Histological subtype (Lauren classification)		
Intestinal	145 (44%)	135 (40%)
Diffuse	115 (35%)	133 (40%)
Mixed	21 (6%)	14 (4%)
Unknown or not available	49 (15%)	53 (16%)
Primary tumour present	209 (63%)	209 (62%)
Number of metastatic sites		
0–2	209 (63%)	232 (69%)
≥3	121 (37%)	103 (31%)
Peritoneal metastases	163 (49%)	152 (45%)
Presence of ascites		
Yes	130 (39%)	107 (32%)
No	200 (61%)	228 (68%)
Weight loss (past 3 months)		
<10%	277 (84%)	286 (85%)
≥10%	53 (16%)	47 (14%)
Previous treatment		
Triplet: platinum and fluoropyrimidine with anthracycline	76 (23%)	87 (26%)
Doublet: platinum and fluoropyrimidine	253 (77%)	246 (73%)
HER2, EGFR, or other	31 (9%)	26 (8%)
Previous surgery for gastric cancer		
Yes	133 (40%)	126 (38%)
Total gastrectomy	52 (16%)	65 (19%)
Partial gastrectomy	80 (24%)	59 (18%)
Other	1 (<1%)	2 (<1%)

Data are number (%) unless otherwise indicated. ECOG=Eastern Cooperative Oncology Group. *By self-report. †Region 1: Europe, Israel, Australia, and the USA; region 2: Argentina, Brazil, Chile, and Mexico; and region 3: Japan, South Korea, Hong Kong, Singapore, and Taiwan. ‡As reported in the interactive voice response system.

Table 1: **Baseline characteristics of patients and their tumours in the intention-to-treat population**

three patients, who had protocol violations, two patients in the placebo plus paclitaxel group had received a platinum-based and fluoropyrimidine-based therapy in the neoadjuvant and adjuvant setting and a fluoropyrimidine-based therapy containing irinotecan and fluorouracil in the first-line setting before enrolling on this study, and one patient in the placebo plus paclitaxel group had received fluoropyrimidine monotherapy in the first-line setting and a fluoropyrimidine and platinum combination in the second-line setting. About two-thirds of the patients had disease progression while still on first-line therapy (table 1). Additionally, a large proportion of patients had other poor prognostic factors including poorly differentiated tumours, disease progression within 6 months after the start of the previous therapy, at least three metastatic sites, presence of primary tumour, peritoneal metastases, or presence of ascites (table 1).

There were 256 deaths in the ramucirumab plus paclitaxel group, and 260 in the placebo plus paclitaxel group (figure 2A). Overall survival with ramucirumab plus paclitaxel was significantly longer than with placebo plus paclitaxel (median 9·6 months [95% CI 8·5–10·8] *vs* 7·4 months [95% CI 6·3–8·4], stratified HR 0·807 [95% CI 0·678–0·962]; p=0·017; figure 2A). 6-month overall survival was 72% (95% CI 66–76) in the ramucirumab plus paclitaxel group, and 57% (95% CI 51–62) in the placebo plus paclitaxel group; 12-month overall survival was 40% (95% CI 35–45) and 30% (95% CI 25–35), respectively (figure 2A).

Overall survival was significantly increased in the ramucirumab plus paclitaxel group compared with the placebo and paclitaxel group (see figure 3 for subgroup analyses).

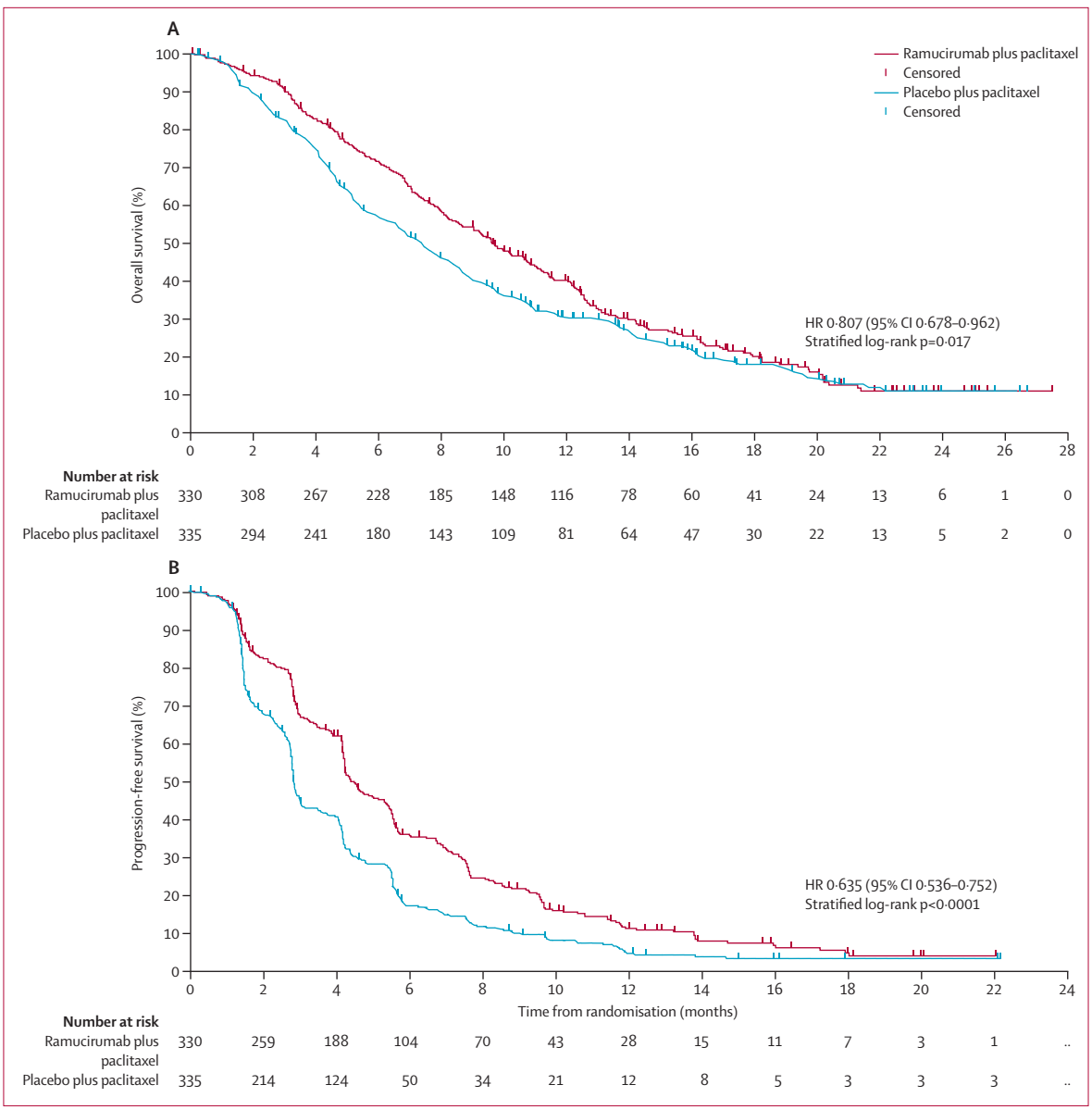

Figure 2: **Kaplan-Meier curves of overall survival (A) and progression-free survival (B)**
HR=hazard ratio.

We did a multivariate analysis using the stepwise Cox model with inclusion of all prespecified factors, and identified seven significant independent predictors for improved survival: region 3, ECOG performance status 0, weight loss of less 10%, up to two metastatic sites, absence of ascites, well or moderately differentiated tumour, and previous gastrectomy. After adjustment for these factors, the HR for overall survival with ramucirumab plus paclitaxel compared with placebo plus paclitaxel was 0·745 (95% CI 0·626–0·888 (p=0·0010; appendix). ECOG performance status, region, and presence of ascites were the strongest predictors for survival (appendix). Slight imbalances in ECOG performance status and ascites between the treatment groups (table 1 and appendix) might have contributed to the difference between the adjusted and unadjusted estimates.

Median progression-free survival with ramucirumab plus paclitaxel was significantly longer than with placebo plus paclitaxel (4·4 months [95% CI 4·2–5·3] *vs* 2·9 months [2·8–3·0]; stratified HR 0·635, [95% CI 0·536–0·752]; p<0·0001; figure 2B). 6-month progression-free survival was 36% (95% CI 31–41) in the ramucirumab plus paclitaxel group and 17% (13–22) in the placebo plus paclitaxel group; 9-month progression-free survival was 22% (95% CI 17–27) and 10% (7–14; figure 2B). Progression-free survival for the ramucirumab plus paclitaxel was

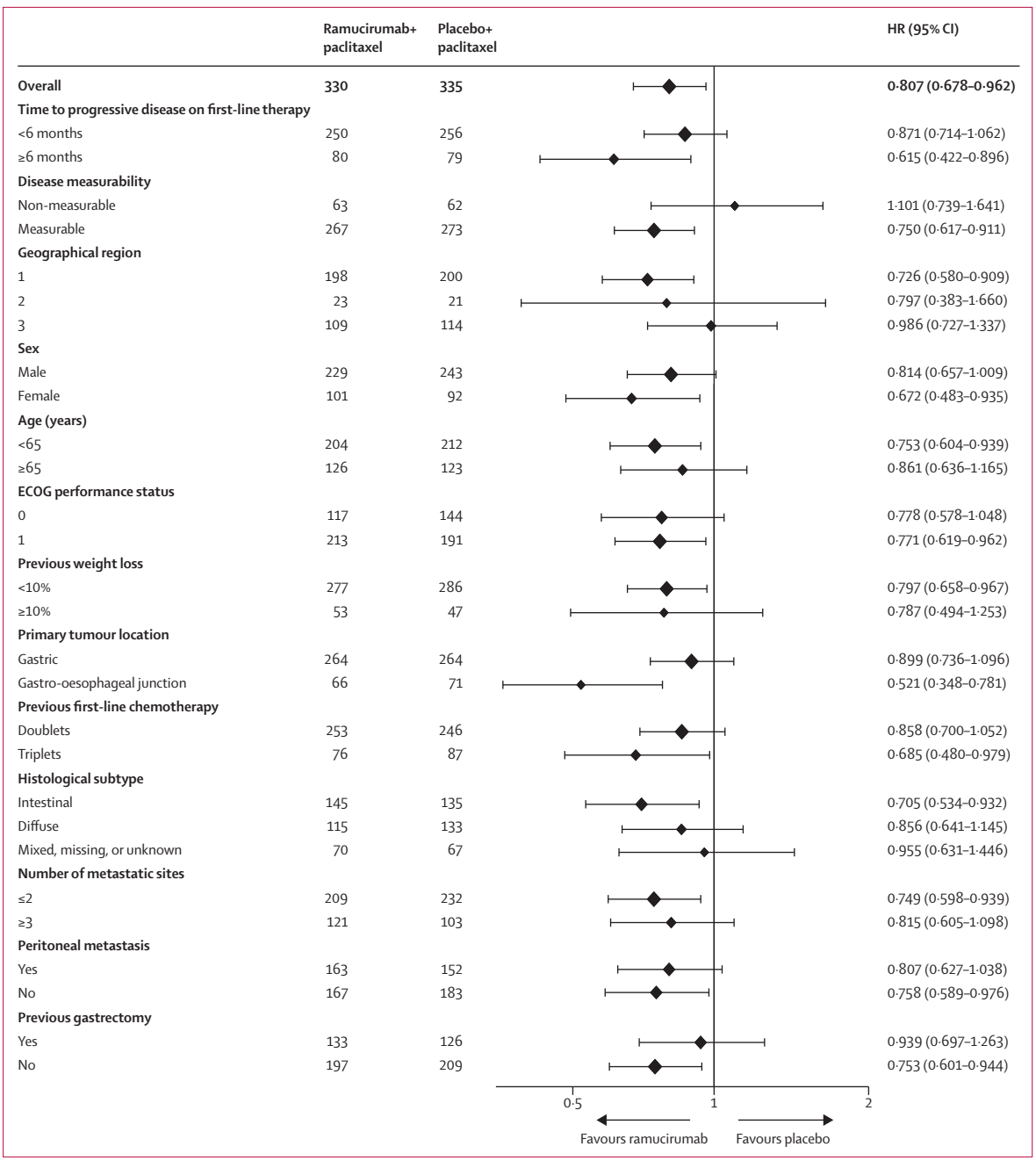

Figure 3: **Forest plot for subgroup univariate analyses of overall survival**
Data are stratified HR (95% CI). The size of the diamonds is proportional to the size of the subgroup. Geographic regions are defined as region 1: Europe, Israel, Australia, and the USA; region 2: Argentina, Brazil, Chile, and Mexico; and region 3: Japan, South Korea, Hong Kong, Singapore, and Taiwan. ECOG=Eastern Cooperative Oncology Group. HR=hazard ratio.

longer than for the placebo plus paclitaxel group in most subgroups (figure 4). Progression-free survival was significantly increased in the ramucirumab plus paclitaxel group compared with the placebo plus paclitaxel group after adjustment for significant baseline factors (adjusted HR 0·599; [95% CI 0·506–0·708]; p<0·0001; appendix).

A significantly greater proportion of patients achieved an objective response in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group (92 [28%, 95% CI 23–33] of 330 *vs* 54 [16%, 13–20] of 335, respectively; p=0·0001, table 2). A significantly greater proportion of patients also achieved disease control in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group (264 [80%, 95% CI 75–84] *vs* 213 [64%, 58–69], respectively; p<0·0001). The median duration of response was longer in the ramucirumab

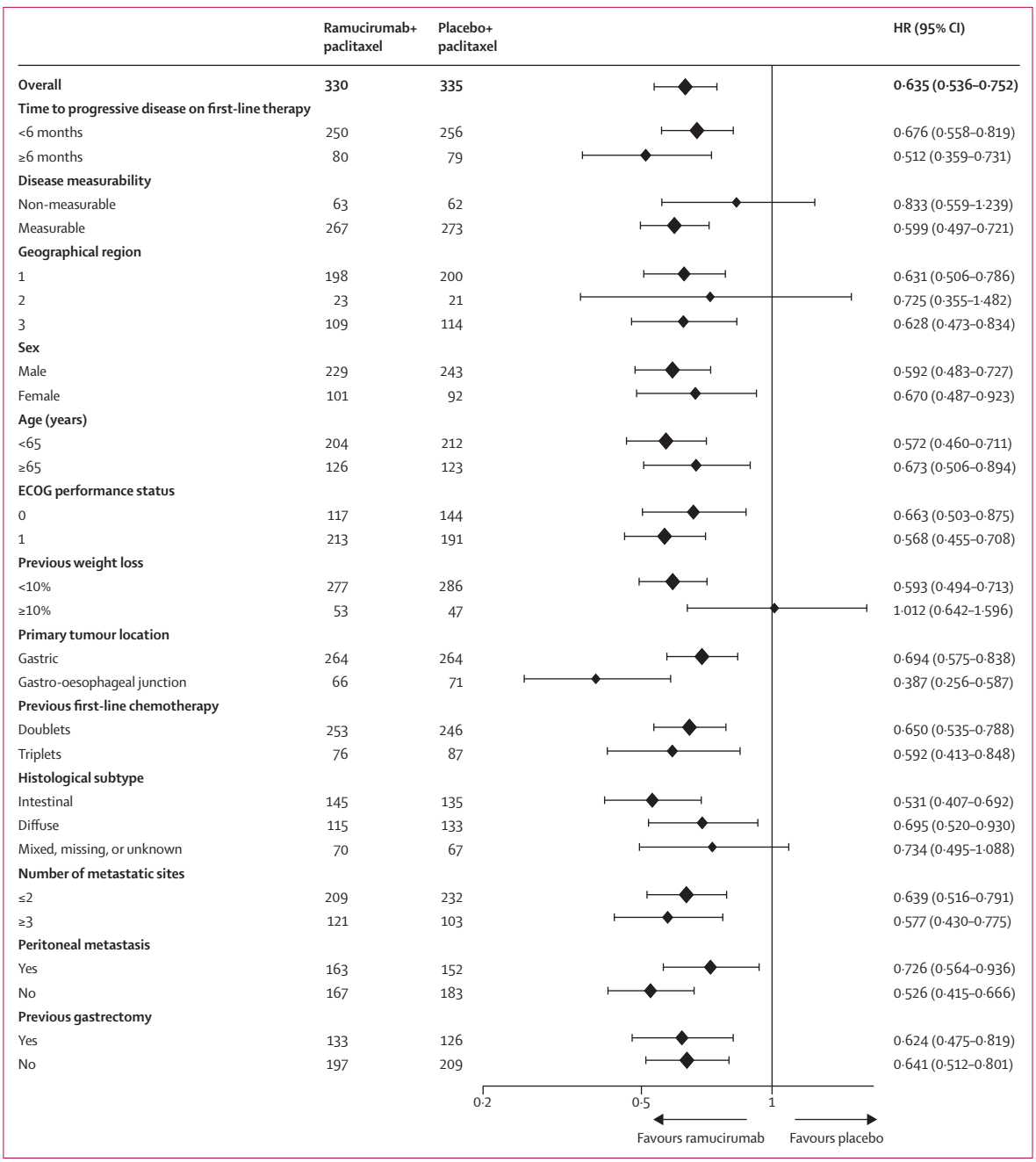

Figure 4: **Forest plots for subgroup univariate analyses of progression-free survival**
Data are stratified HR (95% CI). The size of the diamonds is proportional to the size of the subgroup. Geographic regions are defined as region 1: Europe, Israel, Australia, and the USA; region 2: Argentina, Brazil, Chile, and Mexico; and region 3: Japan, South Korea, Hong Kong, Singapore, and Taiwan. ECOG=Eastern Cooperative Oncology Group. HR=hazard ratio.

plus paclitaxel group than in the placebo plus paclitaxel group (4·4 months [IQR 2·8–7·5], *vs* 2·8 months [1·4–4·4], respectively).

Table 3 summarises overall survival, progression-free survival, and the proportion of patients achieving an objective response by geographic regions, comparing Asian with non-Asian patients. Overall survival in the ramucirumab plus paclitaxel group compared with placebo plus paclitaxel was not significantly increased for patients in region 3 compared with those in regions 1 and 2.

Baseline and end-of-treatment results for global quality of life from the QLQ-C30 and index scores from the EQ-5D-3L were similar in the treatment groups (table 4). Further details for quality of life will be published separately.

Median duration of treatment with ramucirumab was 18·0 weeks (IQR 10·0–31·1) in the ramucirumab plus paclitaxel group, and 12·0 weeks (6·4–20·0) with placebo in the placebo plus paclitaxel group. Median relative dose intensity of ramucirumab was similar to placebo (99% [IQR 94–101] *vs* 100% [97–101]) and was similar for paclitaxel in both groups (88% [IQR 72–97] *vs* 93% [85–99]). Dose reductions of ramucirumab occurred in 16 (5%) of 327 patients in the ramucirumab plus paclitaxel group, and of placebo in three (<1%) of 329 patients in the placebo plus paclitaxel group. Paclitaxel dose reductions occurred in 78 (24%) patients in the ramucirumab plus paclitaxel group, and in 24 (7%) patients in the placebo plus paclitaxel group. Median cumulative doses and number of infusions are provided in the appendix.

Disease progression was the most common reason for treatment discontinuation in both treatment groups (236 [72%] of 330 in the ramucirumab plus paclitaxel group *vs* 255 [76%] of 335 in the placebo plus paclitaxel group); 39 (12%) and 38 (11%) patients, respectively, discontinued treatment because of adverse events (appendix). One patient (who was randomly assigned to placebo but received ramucirumab) was unmasked to the investigator before surgery because of the occurrence of serious adverse events (sepsis and intestinal occlusion). After the discontinuation of study drug, the number of patients receiving systemic anti-neoplastic treatment was similar in both groups (appendix). Of note, a higher percentage of patients in region 3 received treatment after discontinuation of the study drug than in regions 1 or 2 (appendix).

Nine randomly assigned patients did not receive study medication and were excluded from the safety analyses (figure 1). Hence, 656 patients were included in the safety analyses. The patient who was randomly assigned to the placebo plus paclitaxel group but erroneously received ramucirumab instead of placebo discontinued treatment after one infusion. This patient is included in the intention-to-treat population (as randomly assigned) in the placebo plus paclitaxel group and is included in the safety population (as treated) in the ramucirumab plus paclitaxel group. Consequently, the safety population consisted of 327 patients in the ramucirumab plus paclitaxel group and 329 patients in the placebo plus paclitaxel group.

The incidence of grade 3 or 4 adverse events was higher in the ramucirumab plus paclitaxel group, including grade 3 or 4 neutropenia, leucopenia, and grade 3 hypertension, abdominal pain, and fatigue (table 5). All grade 3–5 adverse events are listed in the appendix. Although the incidence of grade 3 or 4 neutropenia was higher in the ramucirumab plus paclitaxel group, the incidence of grade 3 or greater febrile neutropenia was similar in both groups (ten [3%] *vs* eight [2%]). Neuropathy, of all grades, was more common in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group (table 5), and was associated with a higher cumulative paclitaxel dose (appendix).

	Ramucirumab plus paclitaxel (N=330)	Placebo plus paclitaxel (N=335)
Best overall response		
Complete response	2 (<1%)	1 (<1%)
Partial response	90 (27%)	53 (16%)
Stable disease	172 (52%)	159 (47%)
Progressive disease	43 (13%)	83 (25%)
Not evaluable or not assessed	23 (7%)	39 (12%)

Data are number (%) or number (%; 95% CI), unless otherwise indicated.

Table 2: **Best overall response**

	Ramucirumab plus paclitaxel	Placebo plus paclitaxel	Hazard ratio (95% CI)	Odds ratio (95% CI)
Median overall survival				
Regions 1 (n=398) and 2 (n=44)	8·5 months (7·4–9·8)	5·9 months (5·2–7·1)	0·732 (0·591–0·907)	
Region 3 (n=223)	12·1 months (10·0–13·3)	10·5 months (7·8–14·1)	0·986 (0·727–1·337)	
Median progression-free survival				
Region 1 (n=398) and 2 (n=44)	4·2 months (3·9–4·9)	2·9 months (2·6–3·5)	0·639 (0·518–0·788)	
Region 3 (n=223)	5·5 months (4·2–5·7)	2·8 months (2·8–4·1)	0·628 (0·473–0·834)	
Proportion of patients achieving an objective response				
Regions 1 (n=398) and 2 (n=44)	55 (25%)	31 (14%)		2·087 (1·278–3·409)
Region 3 (n=223)	37 (34%)	23 (20%)		2·235 (1·177–4·244)

Data are median (95% CI) or number (%), unless otherwise indicated. Region 1=Europe, Israel, Australia, USA. Region 2=Argentina, Brazil, Chile, and Mexico. A pooled analysis is presented for regions 1 and 2 because of the similarity of the patient populations in these two regions and the small sample size in region 2. Region 3=Japan, South Korea, Hong Kong, Singapore, and Taiwan.

Table 3: **Efficacy by geographic region**

	Ramucirumab plus paclitaxel (n=330)	Placebo plus paclitaxel (n=335)
EORTC QLQ-C30		
Patients completing baseline QLQ-C30	322 (98%)	328 (98%)
Baseline global quality-of-life score*	61·5 (22·0)	58·0 (22·0)
Patients completing baseline plus post-baseline QLQ-C30	287 (87%)	273 (81%)
Patients completing end-of-treatment QLQ-C30	211 (64%)	204 (61%)
End-of-treatment global quality-of-life score*	49·0 (23·0)	48·3 (23·9)
EQ-5D		
Patients completing baseline EQ-5D	323 (98%)	328 (98%)
Baseline index score†	0·75 (0·22)	0·75 (0·24)
Patients completing baseline plus post-baseline EQ-5D	287 (87%)	274 (82%)
Patients completing end-of-treatment EQ-5D	211 (64%)	206 (61%)
End-of-treatment index score†	0·61 (0·32)	0·60 (0·35)

Data are number (%) or mean (SD). EORTC QLQ-C30=European Organisation for Research and Treatment of Cancer quality-of-life questionnaire. EQ-5D=EuroQol five-dimension health questionnaire. *Based on a 100-point scale, with a higher score representing better quality of life. †Based on a –0·59 to 1 scale, with 1 representing perfect health.

Table 4: **Quality of life QLQ-C30 and EQ-5D mean scores at baseline and end of treatment**

	Ramucirumab plus paclitaxel (n=327)				Placebo plus paclitaxel (n=329)			
	Grades 1–2	Grade 3	Grade 4	Grade 5	Grades 1–2	Grade 3	Grade 4	Grade 5
Any patients with a treatment-emergent adverse event	57 (17%)	155 (47%)	73 (22%)	39 (12%)	116 (35%)	128 (39%)	27 (8%)	51 (16%)
Non-haematological adverse events								
Fatigue*	147 (45%)	39 (12%)	0	0	126 (38%)	18 (5%)	0	0
Neuropathy*	123 (38%)	27 (8%)	0	0	104 (32%)	15 (5%)	0	0
Decreased appetite	121 (37%)	10 (3%)	0	0	92 (28%)	13 (4%)	0	0
Abdominal pain*	98 (30%)	20 (6%)	0	0	87 (26%)	10 (3%)	1 (<1%)	0
Nausea	109 (33%)	5 (2%)	1 (<1%)	0	100 (30%)	8 (2%)	0	0
Alopecia	107 (33%)	0	0	0	126 (38%)	1 (<1%)	0	0
Diarrhoea	94 (29%)	12 (4%)	0	0	71 (22%)	4 (1%)	1 (<1%)	0
Epistaxis	100 (31%)	0	0	0	23 (7%)	0	0	0
Vomiting	78 (24%)	9 (3%)	1 (<1%)	0	56 (17%)	12 (4%)	0	0
Peripheral oedema	77 (24%)	5 (2%)	0	0	43 (13%)	2 (<1%)	0	0
Hypertension	32 (10%)	46 (14%)	0	0	8 (2%)	8 (2%)	0	0
Constipation	70 (21%)	0	0	0	69 (21%)	2 (<1%)	0	0
Stomatitis	62 (19%)	2 (<1%)	0	0	22 (7%)	2 (<1%)	0	0
Pyrexia	56 (17%)	3 (<1%)	0	0	36 (11%)	1 (<1%)	0	0
Proteinuria	50 (15%)	4 (1%)	0	0	20 (6%)	0	0	0
Malignant neoplasm progression	5 (2%)	16 (5%)	4 (1%)	27 (8%)	1 (<1%)	24 (7%)	1 (<1%)	34 (10%)
Weight decreased	39 (12%)	6 (2%)	0	0	45 (14%)	4 (1%)	0	0
Dyspnoea	34 (10%)	8 (2%)	0	0	29 (9%)	2 (<1%)	0	0
Rash*	42 (13%)	0	0	0	31 (9%)	0	0	0
Cough	40 (12%)	0	0	0	25 (8%)	0	0	0
Back pain	35 (11%)	4 (1%)	0	0	35 (11%)	5 (2%)	0	0
Hypoalbuminaemia*	32 (10%)	4 (1%)	0	0	13 (4%)	2 (<1%)	0	1 (<1%)
Myalgia	34 (10%)	0	0	0	32 (10%)	1 (<1%)	0	0
Ascites	21 (6%)	11 (3%)	1 (<1%)	0	14 (4%)	13 (4%)	0	0
Headache	32 (10%)	0	0	0	21 (6%)	1 (<1%)	0	0
Haematological adverse events								
Neutropenia*	45 (14%)	71 (22%)	62 (19%)	0	40 (12%)	51 (16%)	11 (3%)	0
Anaemia*	84 (26%)	30 (9%)	0	0	85 (26%)	31 (9%)	3 (<1%)	0
Leucopenia*	54 (17%)	52 (16%)	5 (2%)	0	47 (14%)	19 (6%)	3 (<1%)	0
Thrombocytopenia*	38 (12%)	5 (2%)	0	0	14 (4%)	6 (2%)	0	0

Data are number (%), unless otherwise stated. *Consolidated adverse event category comprising synonymous MEdDRA preferred terms.

Table 5: **Treatment-emergent adverse events occurring in at least 10% of patients on ramucirumab plus paclitaxel, irrespective of causality**

Grade 3 adverse events that were potentially associated with the VEGF pathway—and thus were of special interest—that were more common in the ramucirumab plus paclitaxel group included hypertension, proteinuria, and bleeding or haemorrhage (table 6). The incidences of grade 4 and 5 adverse events of special interest were low in both groups, with no grade 4 or 5 hypertension, a similar incidence of gastrointestinal haemorrhage, and a higher incidence of gastrointestinal perforation in the ramucirumab plus paclitaxel group than the placebo plus paclitaxel group (table 6).

Similar numbers of patients had at least one serious adverse event (153 [47%] of 327 in the ramucirumab plus paclitaxel group *vs* 139 [42%] of 329 in the placebo plus paclitaxel group), or treatment-emergent adverse event leading to death (39 [12%] *vs* 51 [16%], respectively). Six (2%) patients in the ramucirumab plus paclitaxel group had adverse events leading to death with a causal relation to any study drug, which were septic shock; malabsorption; gastrointestinal haemorrhage; death of unknown origin; pulmonary embolism; and sepsis. Five (2%) patients in the placebo plus paclitaxel group had adverse events leading to death with a causal relation to any study drug, which were acute renal failure; cardiac failure; febrile neutropenia, septic shock, and pulmonary embolism; pulmonary embolism; and cerebral haemorrhage.

Serum samples for detection of anti-ramucirumab antibodies were available for 320 (98%) of 327 patients receiving ramucirumab plus paclitaxel and 323 (98%) of 329 patients receiving placebo plus paclitaxel. Five (2%) patients receiving ramucirumab plus paclitaxel and one

(<1%) patient receiving placebo plus paclitaxel had a positive response. No patients developed neutralising antibodies.

Discussion

Ramucirumab plus paclitaxel significantly increased overall survival compared with placebo plus paclitaxel in patients with advanced gastric or gastro-oesophageal junction adenocarcinoma that had progressed after first-line chemotherapy. Patients treated with ramucirumab plus paclitaxel also had significantly longer progression-free survival, and a higher proportion of patients achieving an overall response and disease control than did those treated with placebo plus paclitaxel. Increased proportions of patients achieving an overall response has been noted previously in trials of anti-angiogenic drugs in combination with chemotherapy, especially when there seems to be an improvement in survival.[25] Results of a preplanned subgroup analysis showed a difference in treatment effect for ramucirumab plus paclitaxel on survival between non-Asian (region 1 and 2) and Asian (region 3) regions. We could speculate that the higher use of post-study discontinuation treatment in Asia (almost 70%) than in the non-Asian regions (almost 40%) attenuated the survival benefit in this region.

Fatigue, diarrhoea, and abdominal pain were some of the most frequently reported non-haematological toxicities in both groups, and were more common in the ramucirumab plus paclitaxel group than the placebo plus paclitaxel group. These events are common in patients with gastric cancer; incidences reported in our trial are in the range of what was previously reported in large phase 3 gastric cancer trials.[9,26–28] Peripheral neuropathy, a typical side-effect of taxanes, was more common in the ramucirumab plus paclitaxel group and, as expected, was associated with a higher cumulative paclitaxel dose. Neutropenia was one of the most frequently reported haematological toxicities in both groups, and had a similar incidence to that reported in other trials of the same paclitaxel dose and schedule.[12,13,17,29] Although severe neutropenia was more frequently reported for ramucirumab plus paclitaxel, the incidence of febrile neutropenia was low and similar in the groups. As expected, hypertension, proteinuria, bleeding (mainly grade 1 or 2 epistaxis), and gastrointestinal perforations, adverse events associated with most anti-angiogenic treatments, were more common in the ramucirumab plus paclitaxel group. Grade 3 hypertension was controlled with antihypertensive medication; grade 4 or greater hypertension was not noted in this study. The incidences of grade 3 bleeding (mainly grade 3 gastro-intestinal haemorrhage) and grade 3 proteinuria were higher in the ramucirumab plus paclitaxel group (table 6). However, nephrotic syndrome was not reported, and grade 4 or 5 bleeding events occurred with a similar incidence in both groups. Grade 3 or greater gastro-intestinal perforation was reported only in the ramucirumab plus paclitaxel group (four patients, including one death). Importantly, the overall higher rate of grade 3 or 4 adverse events in the ramucirumab plus paclitaxel group did not result in a higher number of patients discontinuing, or a higher number of deaths, than in the placebo plus paclitaxel group. We also showed that quality of life was maintained during treatment with ramucirumab.

	Ramucirumab plus paclitaxel (n=327)				Placebo plus paclitaxel (n=329)			
	Grades 1–2	Grade 3	Grade 4	Grade 5	Grades 1–2	Grade 3	Grade 4	Grade 5
Bleeding or haemorrhage	123 (38%)	12 (4%)	1 (<1%)	1 (<1%)	51 (16%)	4 (1%)	2 (<1%)	2 (<1%)
Proteinuria	51 (16%)	4 (1%)	0	0	20 (6%)	0	0	0
Liver injury or failure	39 (12%)	12 (4%)	3 (<1%)	0	28 (9%)	11 (3%)	2 (<1%)	0
Hypertension	34 (10%)	48 (15%)	0	0	10 (3%)	9 (3%)	0	0
Gastrointestinal haemorrhage†	21 (6%)	10 (3%)	1 (<1%)	1 (<1%)	15 (5%)	3 (<1%)	1 (<1%)	1 (<1%)
Infusion-related reaction	17 (5%)	2 (<1%)	0	0	12 (4%)	0	0	0
Renal failure	16 (5%)	4 (1%)	2 (<1%)	0	11 (3%)	0	1 (<1%)	2 (<1%)
Congestive heart failure	6 (2%)	2 (<1%)	0	0	2 (<1%)	1 (<1%)	0	1 (<1%)
Venous thromboembolic events	5 (2%)	7 (2%)	0	1 (<1%)	7 (2%)	8 (2%)	1 (<1%)	2 (<1%)
Arterial thromboembolic events	3 (<1%)	1 (<1%)	2 (<1%)	0	2 (<1%)	2 (<1%)	0	1 (<1%)
Gastrointestinal perforation	0	1 (<1%)	2 (<1%)	1 (<1%)	1 (<1%)	0	0	0

*Pooled adverse-event terms. †Events pooled as gastrointestinal haemorrhage are also pooled as bleeding or haemorrhage.

Table 6: **Adverse events of special interest***

To our knowledge, RAINBOW is the largest trial in second-line gastric cancer, and the first report of a survival benefit with a VEGFR-2 targeted antibody in combination with chemotherapy. REGARD, a randomised phase 3 trial that compared ramucirumab as a single agent with best supportive care, showed a significant median survival benefit of 1·4 months, favouring ramucirumab treatment (median survival 5·2 months [IQR 2·3–9·9]) versus best supportive care (3·8 months [IQR 1·7–7·1]).[30] Other recently published randomised second-line gastric cancer trials showed an increase in median survival of about 1·5 months with single-agent chemotherapy (docetaxel or irinotecan) relative to supportive care.[3–5] By contrast, a trial of everolimus versus placebo in the second-line setting did not significantly extend overall survival,[27] and trials of anti-EGFR therapy in the first-line setting adding panitumumab[31] or cetuximab[28] to chemotherapy have also not significantly extended overall survival or progression-free survival, respectively.

The increased overall survival for patients treated with ramucirumab plus paclitaxel compared with

Panel: **Research in context**

Systematic review

We searched PubMed and the abstracts from major oncology congresses (American Society of Clinical Oncology [ASCO] Annual Meeting and ASCO Gastrointestinal Cancers Symposium, European Society for Medical Oncology [ESMO] Annual Meeting and ESMO World Congress on Gastrointestinal Cancer, International Society of Gastrointestinal Oncology Conference, American Association for Cancer Research-National Cancer Institute-European Organisation for Research and Treatment of Cancer Congress, and European Multidisciplinary Cancer Congress). We used MeSH and full-text search terms for gastric and gastro-oesophageal junction cancer and molecular targeted therapies, limiting our results to English-language articles published in the past 4 years; the last search was done on Dec 31, 2013. For PubMed, the search terms were ("molecular targeted therapy") or ("molecular" and "targeted") and ("therapy" or "therapies") and ("gastric neoplasms" or "gastric cancer") or ("gastric" and "cancer") or ("gastroesophageal neoplasms" or "gastroesophageal cancer") or ("gastroesophageal" and "cancer") and ("2009/01/01"and "2013/11/13"). For conferences, the search terms were "metastatic gastric cancer" or "advanced gastric cancer," manually limited to abstracts on targeted therapies or phase 3 studies. We identified several potential targeted agents (monoclonal antibodies or small-molecule tyrosine-kinase inhibitors) that are being investigated either in synergy with, or instead of, established treatments, including inhibitors of growth factors and their receptors (VEGF, EGFR, insulin-like growth factor, HER2, C-MET), MEK inhibitors, and agents targeting hedgehog protein. We used information from the abstracts and ClinicalTrials.gov to identify the latest stage of clinical developments of these agents in gastric and gastro-oesophageal junction cancer. We have limited our discussion to the agents we believe are most promising and relevant to the patient population on the basis of clinical trial efficacy.

Interpretation

Various oncogenic signalling processes have been implicated in gastric cancer. Findings with the monoclonal antibodies trastuzumab and ramucirumab suggest that the HER2 and VEGF signalling pathways might be valid targets for therapy for gastric cancer. RAINBOW shows that, in patients with progressive gastric or gastro-oesophageal adenocarcinoma after standard first-line therapy, ramucirumab in combination with paclitaxel can significantly increase overall survival compared with placebo plus paclitaxel. Together with an earlier phase 3 trial of single-agent ramucirumab in second-line advanced gastric cancer,[30] the results of these studies support the role of a VEGF receptor-targeted therapy and maybe the addition of a new agent to the treatments and standard of care in this patient population.

placebo plus paclitaxel can be regarded as clinically meaningful in this population. In RAINBOW, the median overall survival in patients from Asia (region 3, most from Japan) was longer than in patients from regions 1 and 2, independent of treatment. In a study by Hironaka and colleagues,[17] patients receiving paclitaxel had a similar median overall survival as patients from Asia in our trial (about 10 months). Likewise, the median overall survival in patients receiving irinotecan or docetaxel in non-Asian studies, and in the non-Asian population receiving placebo plus paclitaxel in our trial (regions 1 and 2) was similar (about 4–5 months).[3–5] Together, these findings might explain the higher median overall survival in the placebo plus paclitaxel group than that in previously reported trials in entirely western populations.

Since RAINBOW showed that second-line therapy can significantly improve survival of patients with advanced gastric cancer, ramucirumab plus paclitaxel could be regarded as a new standard second-line treatment for advanced gastric cancer. Our findings, combined with those of the REGARD trial,[30] validate the role of VEGFR-2 signalling as an important therapeutic target in advanced gastric and gastro-oesophageal junction adenocarcinoma (panel). Analyses are ongoing to identify potential predictive biomarkers for ramucirumab.

Contributors

JDS, ME, PR, DC, AO, and HW conceived and designed the study. HW, EVC, S-CO, GB, YS, SH, NS, OL, T-YK, DC, AO, PR, YK, JA, DF, and KM gathered the data. JDS, ME, RC, KC, AO, and HW analysed and interpreted the data. All authors were involved in the drafting, review, and approval of the manuscript and the decision to submit for publication.

Declaration of interests

HW reports honoraria for lectures and advisory boards. EVC reports grants from Eli Lilly and Company during the conduct of the study, and grants from Eli Lilly outside of the submitted work paid to the University of Leuven. YS reports grants from Eli Lilly and Company during the conduct of the study; grants and personal fees from Taiho; grants and personal fees from Chugai; grants and personal fees from Yakult; grants and personal fees from Pfizer; grants from Sanofi; grants from Novartis; personal fees from Bayer; personal fees from Bristol-Myers Squibb; personal fees from Merck Serono outside of the submitted work. SH reports personal fees and non-financial support from Eli Lilly Japan during the conduct of the study. DC reports grants from Roche; grants from Amgen; grants from Celgene; grants from Sanofi; grants from Merck Serono; grants from Novartis; and grants from AstraZeneca outside of the submitted work. PR reports grants and personal fees from Eli Lilly and Company during the conduct of the study; grants from Bayer; grants and personal fees from Sanofi; personal fees from Merck Serono; personal fees from Ipsen; and grants from Novartis outside the submitted work. AO reports personal fees from Taiho; personal fees from Daiichi Sankyo; personal fees from Chugai-Roche; personal fees from Merck Serono; personal fees from Novartis; personal fees from Eisai; and personal fees from Takeda outside the submitted work. YK reports personal fees from Eli Lilly Japan during the conduct of the study and personal fees from Eli Lilly Japan outside of the submitted work. ME, RC, and KC are employees of Eli Lilly and Company. DF reports non-financial support from Eli Lilly and Company during the conduct of the study; personal fees from Eli Lilly and Company outside of the submitted work; and is currently an employee of Eli Lilly and Company. JDS was an employee of the sponsor during the design, conduct, and analysis of the study. The other authors declare no competing interests.

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